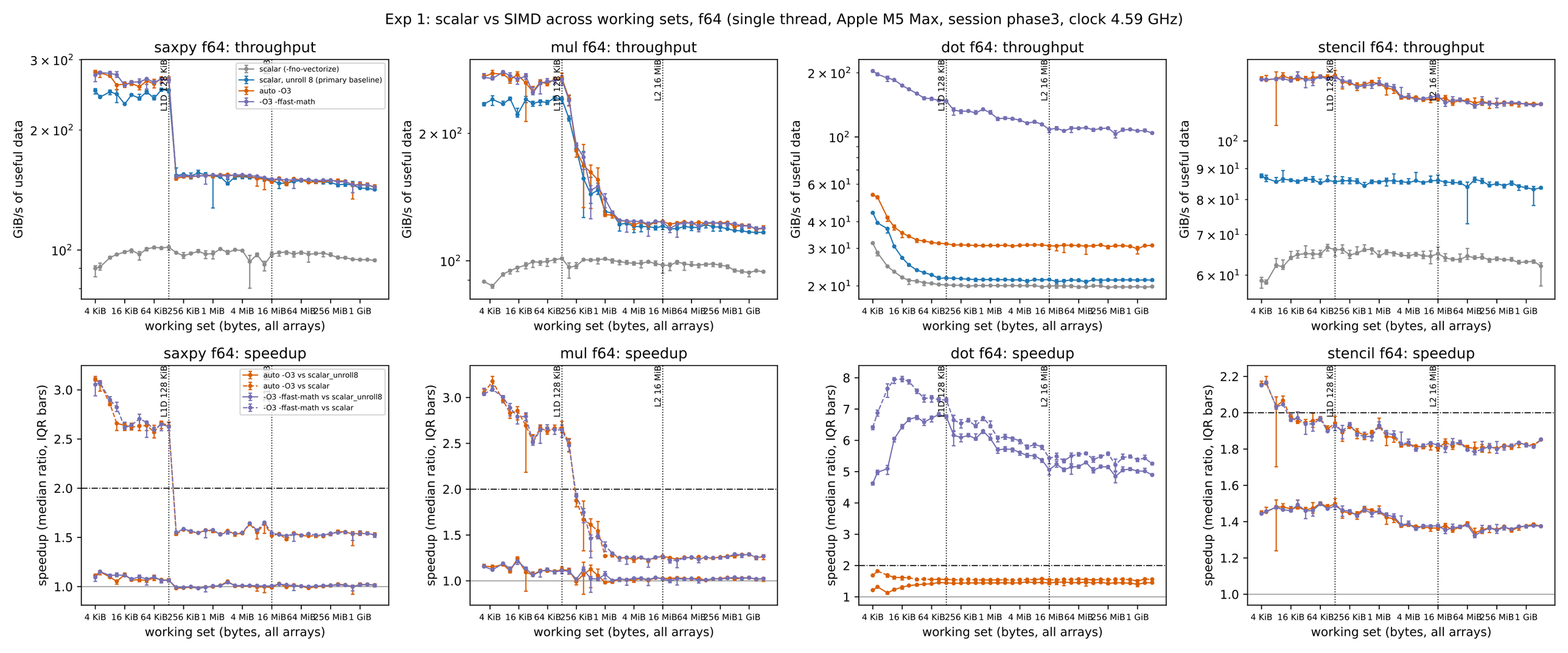

Exp 1: scalar vs SIMD across working sets, f64 (single thread, Apple M5 Max, session phase3, clock 4.59 GHz)
saxpy f64: throughput
mul f64: throughput
dot f64: throughput
stencil f64: throughput
GiB/s of useful data
scalar (-fno-vectorize)
scalar, unroll 8 (primary baseline)
auto -O3
-O3 -ffast-math
L1D 128 KiB
L2 16 MiB
4 KiB 16 KiB 64 KiB 256 KiB 1 MiB 4 MiB 16 MiB 64 MiB 256 MiB 1 GiB
working set (bytes, all arrays)
saxpy f64: speedup
mul f64: speedup
dot f64: speedup
stencil f64: speedup
speedup (median ratio, IQR bars)
auto -O3 vs scalar_unroll8
auto -O3 vs scalar
-O3 -ffast-math vs scalar_unroll8
-O3 -ffast-math vs scalar